High-Risk Lung Cancer Patient Classification Using Machine Learning

Detailed Project Report

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Abstract

This report presents a detailed machine learning project for classifying high-risk lung cancer patients using clinical, demographic, laboratory, and tumor-related features. The project includes data preprocessing, target variable creation, exploratory data analysis, model training, model evaluation, class imbalance handling with SMOTE, feature importance analysis, feature selection, training time comparison, ROC curve analysis, and a graphical user interface prototype. Five supervised machine learning algorithms were tested: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Gradient Boosting. The best-performing models were Logistic Regression, Random Forest, and Gradient Boosting, each reaching approximately 0.71 accuracy. However, ROC-AUC results close to 0.50 show that the models have limited class separation ability. Therefore, this system should be considered as a machine learning prototype and decision-support experiment rather than a clinical diagnosis tool.

Contents

1 Introduction

Lung cancer is one of the most serious cancer types because it is often diagnosed at advanced stages and may require urgent medical intervention. In healthcare, identifying high-risk patients is important because these patients may need closer monitoring, faster treatment planning, and more detailed clinical evaluation.

The main purpose of this project is to develop a machine learning-based classification system that predicts whether a lung cancer patient belongs to the high-risk class or not. Instead of manually analyzing each patient record, machine learning models can process many features at the same time and learn patterns from the dataset.

This project focuses on the complete machine learning pipeline. The work is not limited to model training only. It includes data understanding, exploratory data analysis, preprocessing, class imbalance handling, model comparison, interpretation, and simple deployment through a graphical user interface.

2 Project Objective

The main objectives of this project are:

- To analyze a lung cancer dataset using exploratory data analysis.
- To create a binary target variable named `High_Risk`.
- To preprocess categorical and numerical variables.
- To train multiple supervised machine learning models.
- To compare model performances using accuracy, confusion matrix, ROC curve, and AUC score.
- To handle class imbalance using SMOTE.
- To determine the most important features using feature importance analysis.
- To compare all features with selected top features.
- To build a simple graphical user interface prototype.

3 Dataset Description

The dataset used in this project is named:

`lung_cancer_data.csv`

The dataset contains patient-level information related to lung cancer. It includes demographic features, tumor-related features, treatment-related features, comorbidities, laboratory measurements, and general patient health indicators.

3.1 Main Feature Groups

The dataset contains the following main feature groups:

Table 1: Feature Groups in the Dataset

Feature Group	**Example Features**
Demographic Features	Age, Gender, Ethnicity, Insurance Type
Tumor Features	Tumor Size, Tumor Location, Stage
Clinical History	Smoking History, Smoking Pack Years, Family History
Treatment Information	Treatment Type, Survival Months
Comorbidities	Diabetes, Hypertension, Heart Disease, Chronic Lung Disease, Kidney Disease, Autoimmune Disease
Laboratory Results	Hemoglobin, White Blood Cell Count, Platelet Count, Albumin, LDH, Calcium, Phosphorus, Glucose, Sodium, Potassium
General Condition	Performance Status, Blood Pressure Values

4 Target Variable Creation

The original dataset did not directly contain a ready-made high-risk classification target. Therefore, a new binary target variable named `High_Risk` was created.

The target variable was defined using two important clinical indicators:

- **Performance Status**
- **Cancer Stage**

A patient was labeled as high-risk if:

- Performance Status is greater than or equal to 3, or
- Cancer Stage is 2 or 3.

The mathematical definition is:

$$High\_Risk = \begin{cases} 1, & Performance\_Status \geq 3 \text{ or Stage} \in \{2,3\} \\ 0, & otherwise \end{cases} \tag{1}$$

4.1 Meaning of the Target Classes

Table 2: High_Risk Target Class Meaning

Class	**Meaning**
0	Patient is classified as non-high-risk according to selected clinical rules.
1	Patient is classified as high-risk because of advanced stage or poor performance status.

5 Machine Learning Workflow

The general workflow of the project is shown below.

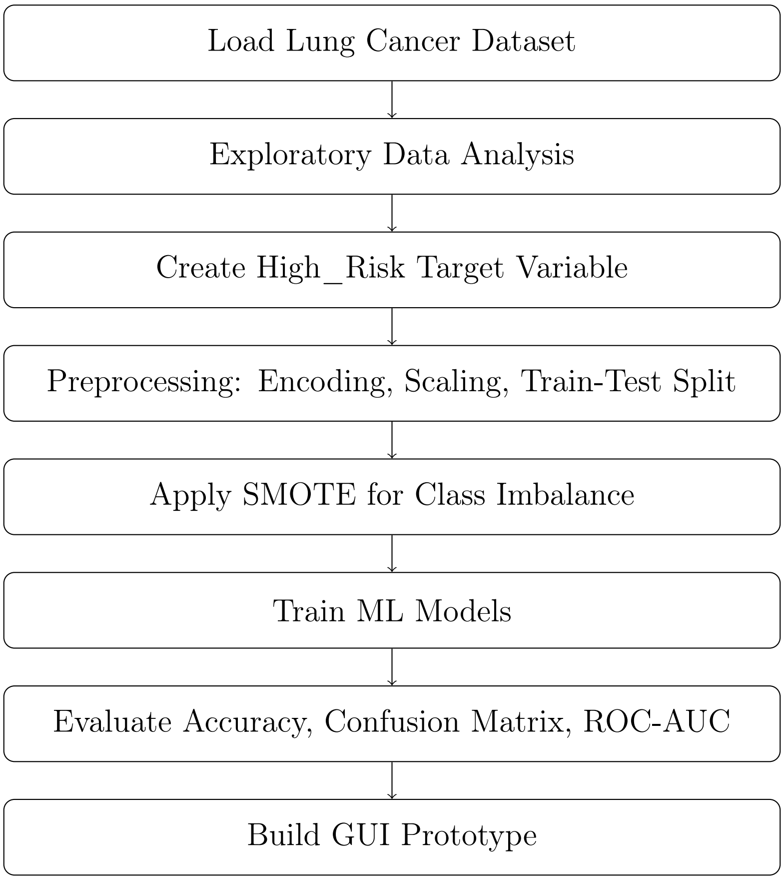

Figure 1: General machine learning workflow of the project.

6 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the dataset structure before model training. This stage helped us understand class distribution, feature ranges, relationships between variables, and possible imbalance problems.

6.1 Target Distribution

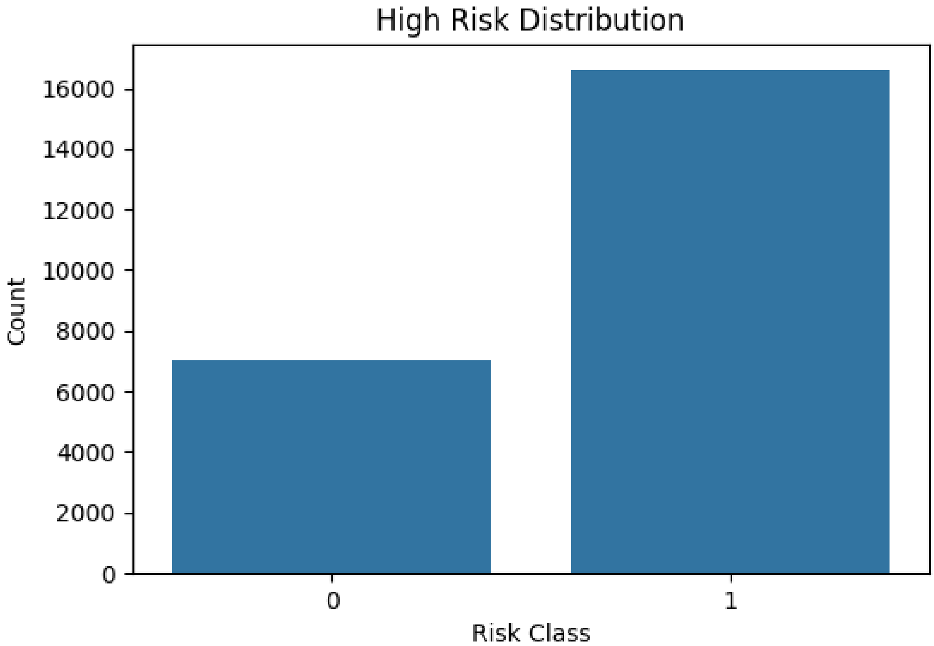

Figure 2: High Risk Distribution

The target distribution shows that class 1, which represents high-risk patients, is more frequent than class 0. This means the dataset is imbalanced. In medical machine learning, imbalance is important because the model may become biased toward the majority class.

6.2 Feature Distribution

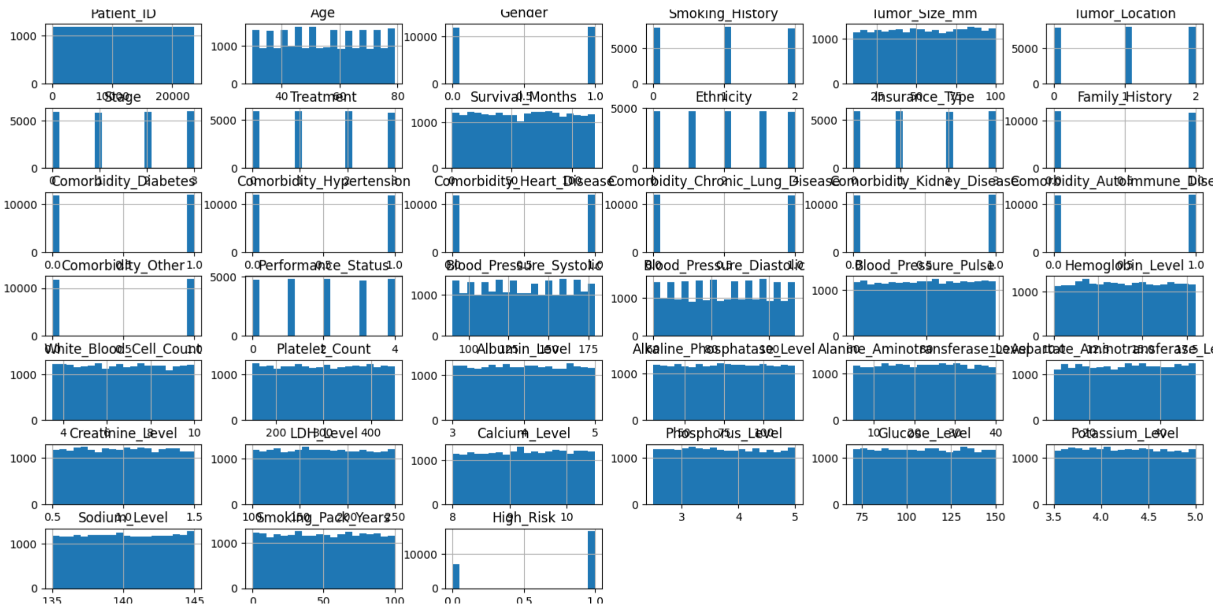

Figure 3: Feature distribution of all dataset columns

The feature distribution graph shows the general distribution of all numerical and encoded categorical variables. Some variables are binary, some are categorical, and some are continuous. This confirms the need for preprocessing before model training.

6.3 Tumor Size Analysis

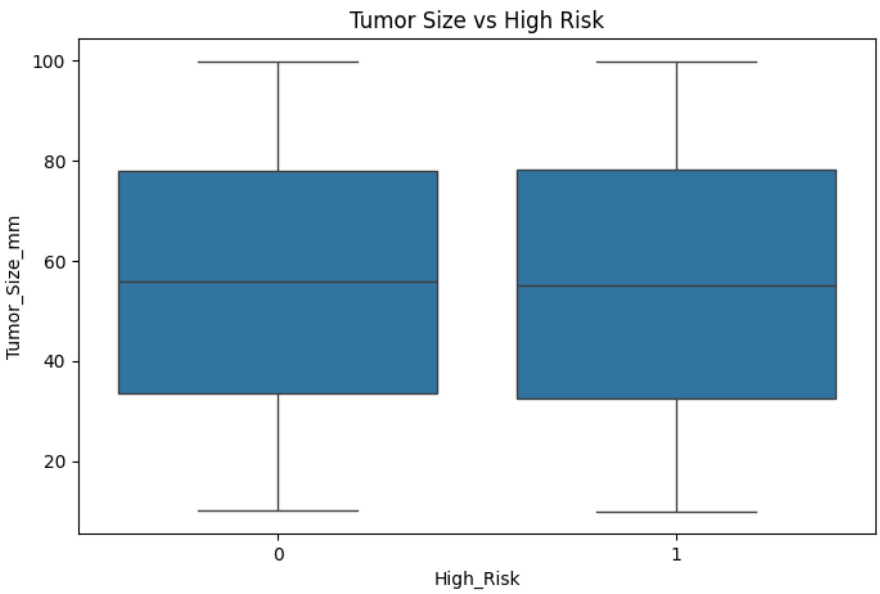

Figure 4: Tumor Size vs High Risk

The tumor size boxplot shows that tumor size distributions for high-risk and non-high-risk patients are similar. This means tumor size alone is not enough to clearly separate the risk classes.

6.4 Correlation Matrix

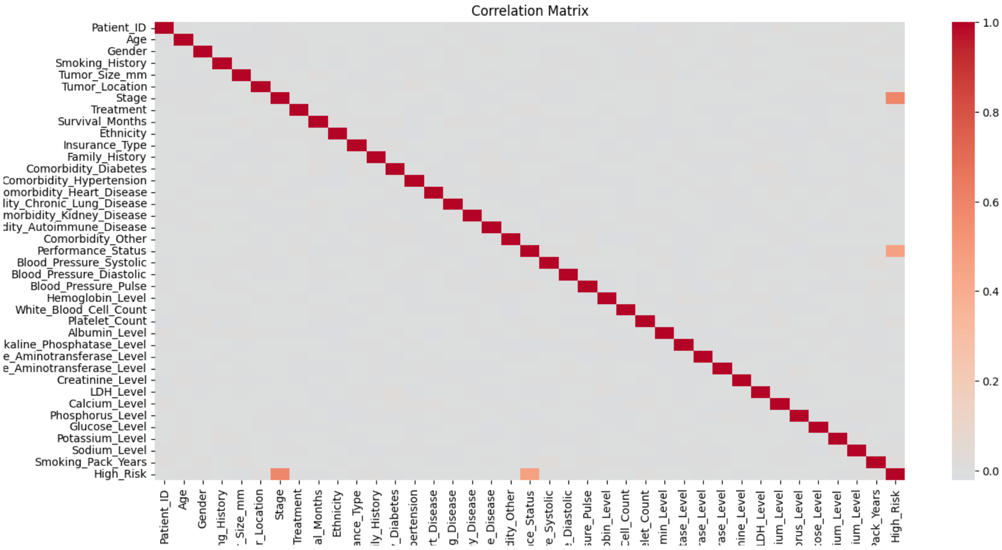

Figure 5: Correlation heatmap of dataset variables

The correlation heatmap was used to examine the relationships between features. The most visible relationship with `High_Risk` is related to `Stage` and `Performance_Status`. This is expected because these variables were used while creating the target variable.

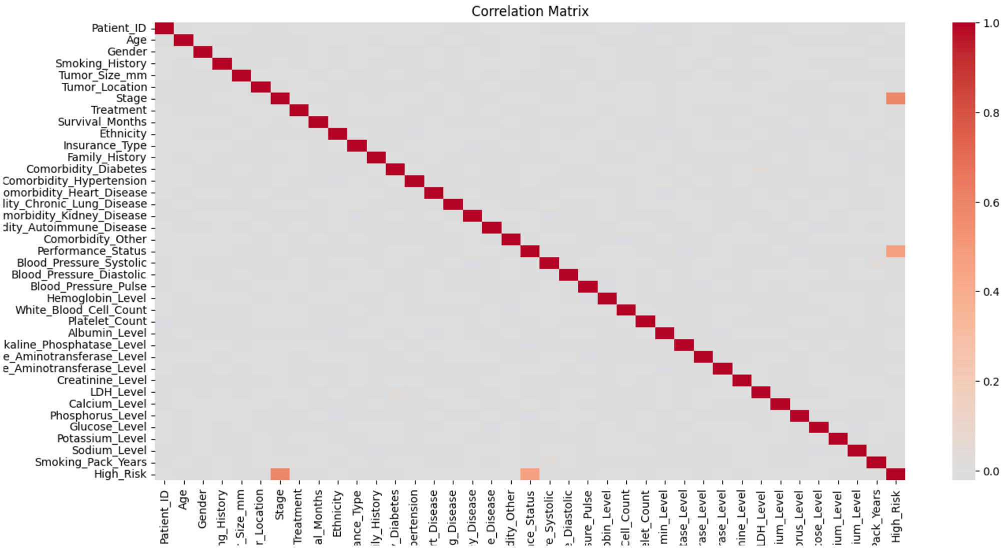

Figure 6: Alternative correlation matrix visualization

7 Data Preprocessing

Data preprocessing is one of the most important parts of the project. Raw data cannot usually be given directly to machine learning models. Therefore, several preprocessing steps were applied.

7.1 Missing Value Control

The dataset was checked for missing values. Missing value analysis helps prevent training errors and unreliable predictions.

7.2 Categorical Encoding

Machine learning models require numerical input. Therefore, categorical columns such as gender, smoking history, tumor location, treatment type, ethnicity, and insurance type were converted into numerical format.

7.3 Feature Scaling

Feature scaling was applied using standardization. This step is especially important for models such as K-Nearest Neighbors because distance-based models are sensitive to feature scale.

7.4 Train-Test Split

The dataset was divided into training and testing subsets. The training set was used to train the models, while the test set was used to evaluate the final performance.

8 Class Imbalance and SMOTE

The target distribution showed that the dataset was imbalanced. In an imbalanced dataset, a model may learn to predict the majority class more frequently. To reduce this problem, SMOTE was applied.

SMOTE means Synthetic Minority Over-sampling Technique. It creates synthetic examples for the minority class instead of simply duplicating existing samples.

Table 3: Purpose of SMOTE

Problem	**SMOTE Solution**
Class imbalance	Generates synthetic samples for the minority class
Model bias	Helps the model learn both classes more fairly
Low minority recall	Improves recognition of minority class samples

9 Models Used in the Project

Five supervised machine learning models were used.

Table 4: Machine Learning Models Used

Model	Explanation
Logistic Regression	A linear classification model used as a baseline.
Decision Tree	A tree-based model that splits data according to feature conditions.
Random Forest	An ensemble model that combines multiple decision trees.
K-Nearest Neighbors	A distance-based model that classifies samples according to nearest neighbors.
Gradient Boosting	An ensemble boosting model that builds trees sequentially to improve errors.

10 Model Accuracy Results

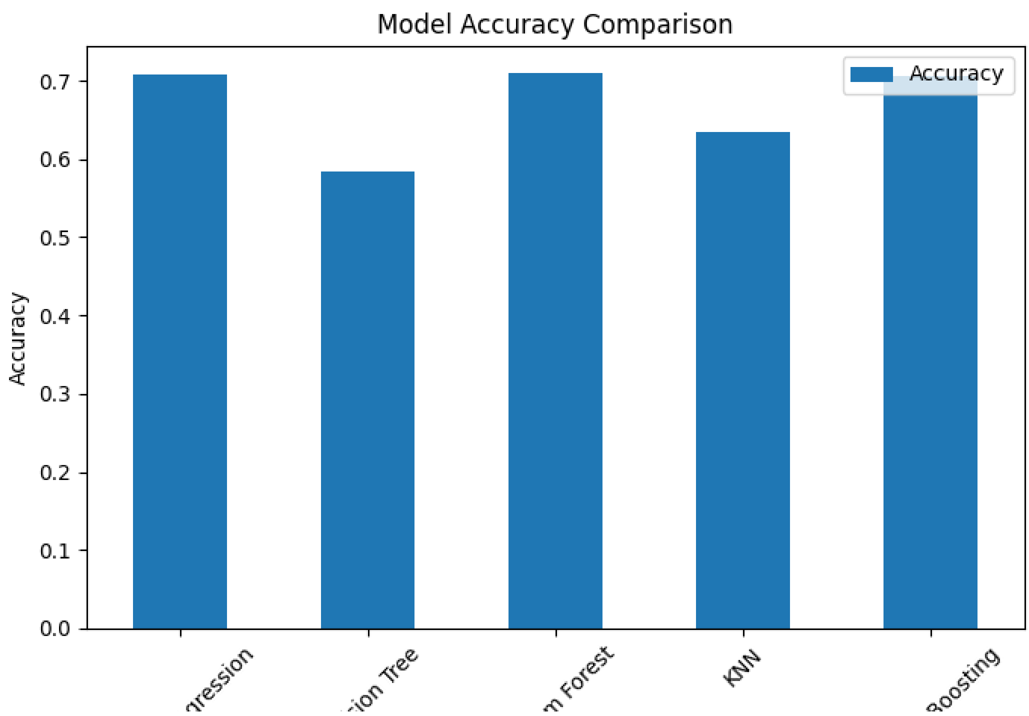

Figure 7: Accuracy comparison of machine learning models

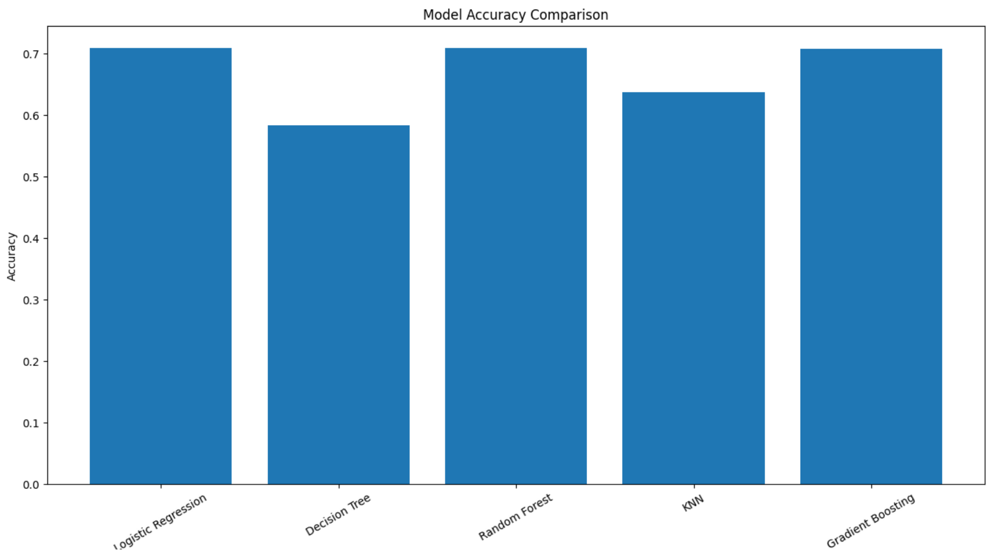

Figure 8: Model accuracy comparison with larger visualization

The model accuracy results are summarized below.

Table 5: Model Accuracy Comparison

Model	**Accuracy**
Logistic Regression	0.71
Decision Tree	0.58
Random Forest	0.71
KNN	0.64
Gradient Boosting	0.71

10.1 Best Model Discussion

According to accuracy values, the best models are:

- Logistic Regression
- Random Forest
- Gradient Boosting

All three models achieved approximately 0.71 accuracy. However, accuracy alone is not enough for medical classification. The ROC curve and confusion matrix results must also be examined.

Random Forest can be considered the most balanced candidate model because:

- It achieved one of the highest accuracy scores.
- It provides feature importance.
- It can model nonlinear relationships.

- It is more interpretable than some complex models.

Therefore, in this project, **Random Forest is selected as the best practical model**, even though Logistic Regression and Gradient Boosting reached similar accuracy.

11 Confusion Matrix Analysis

Confusion matrices were used to understand prediction behavior in more detail.

11.1 General Confusion Matrix

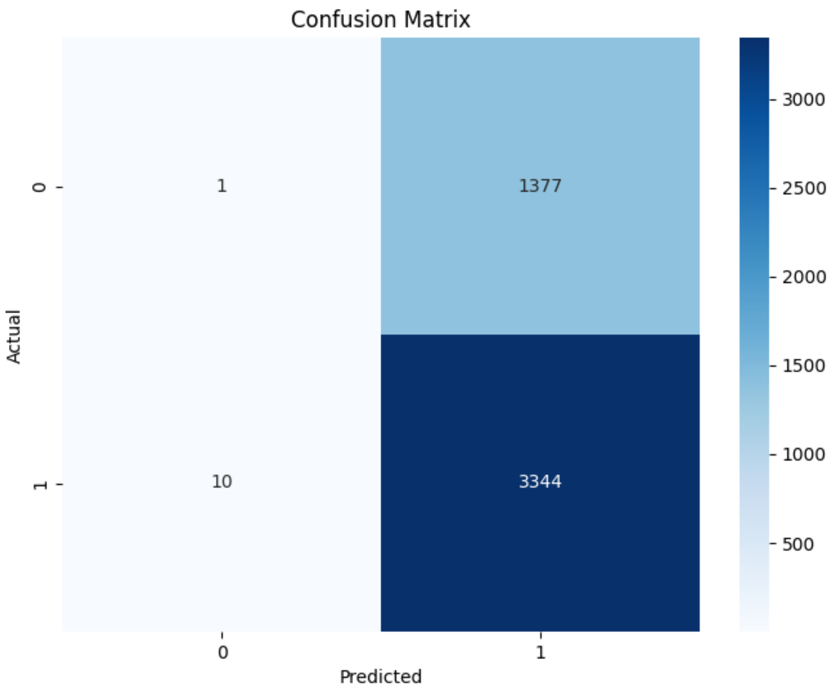

Figure 9: General confusion matrix

11.2 Confusion Matrix Before SMOTE

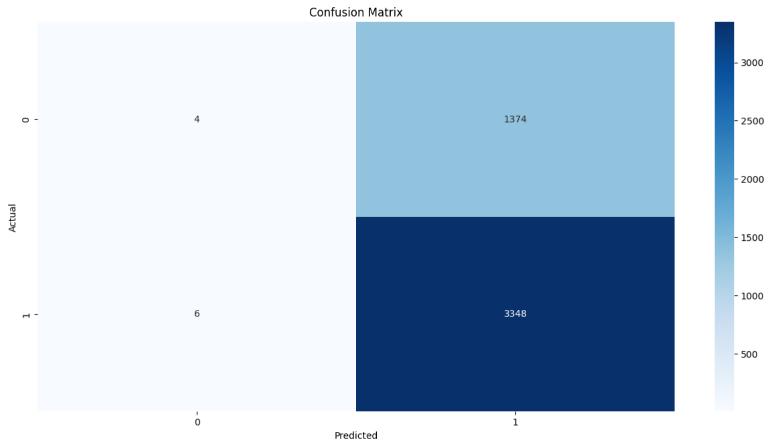

Figure 10: Confusion matrix before SMOTE

Before SMOTE, the model predicted most samples as class 1. This shows that the model was strongly affected by the majority class.

11.3 Confusion Matrix After SMOTE

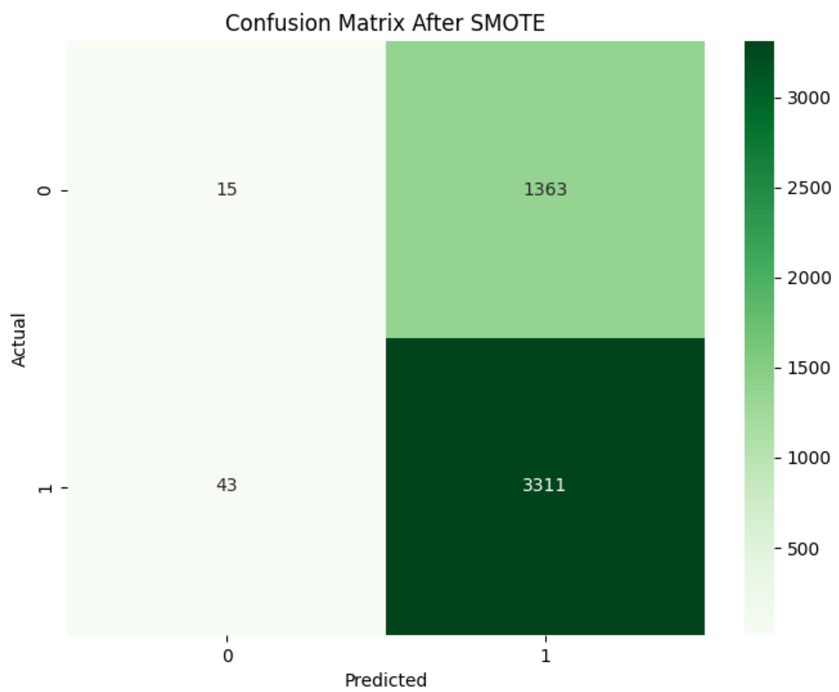

Figure 11: Confusion matrix after SMOTE

After SMOTE, the model showed a small improvement in detecting class 0. However, many class 0 samples were still predicted as class 1. This means that imbalance handling helped, but it did not completely solve the classification problem.

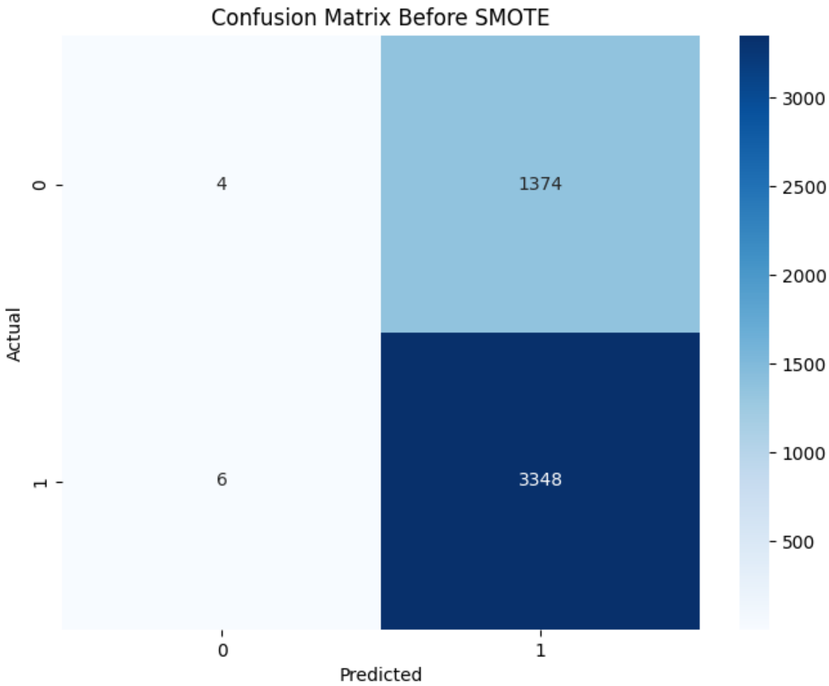

Figure 12: Additional SMOTE confusion matrix result

12 ROC Curve and AUC Analysis

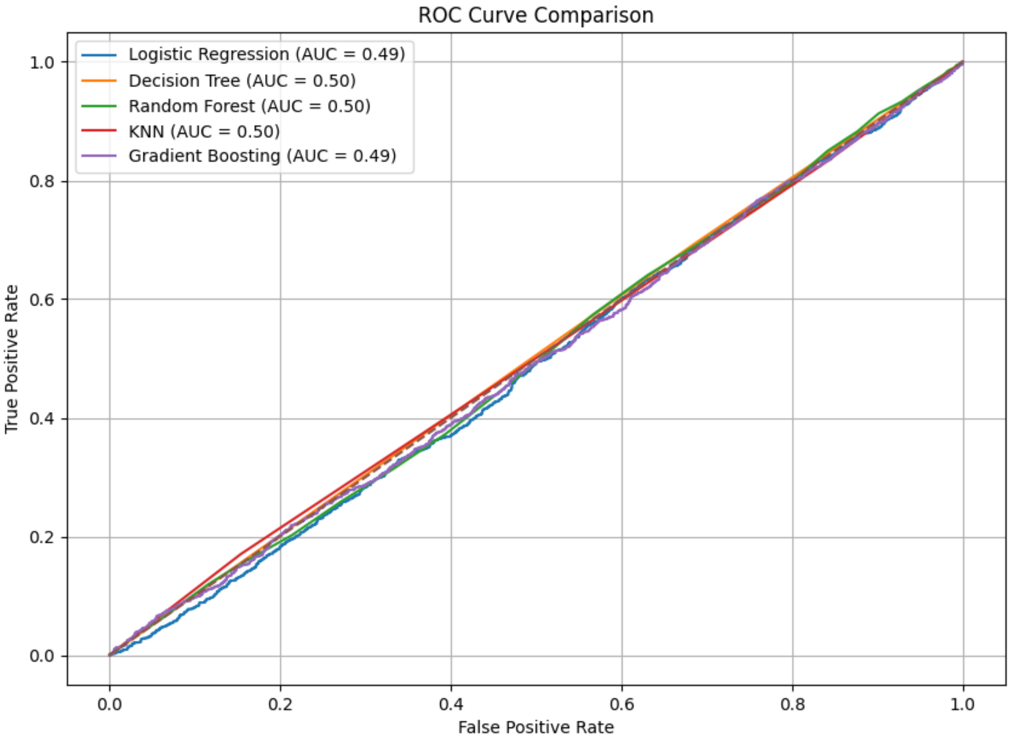

Figure 13: ROC curve comparison of all models

The ROC curve results show that the models have AUC values close to 0.50. This means the models have weak separation ability between high-risk and non-high-risk patients.

This is an important result. Although some models reached 0.71 accuracy, their ROC-AUC values show that the models may mostly learn the dominant class distribution instead of truly separating both classes.

13 Feature Importance Analysis

Feature importance analysis was used to understand which features contributed most to model decisions.

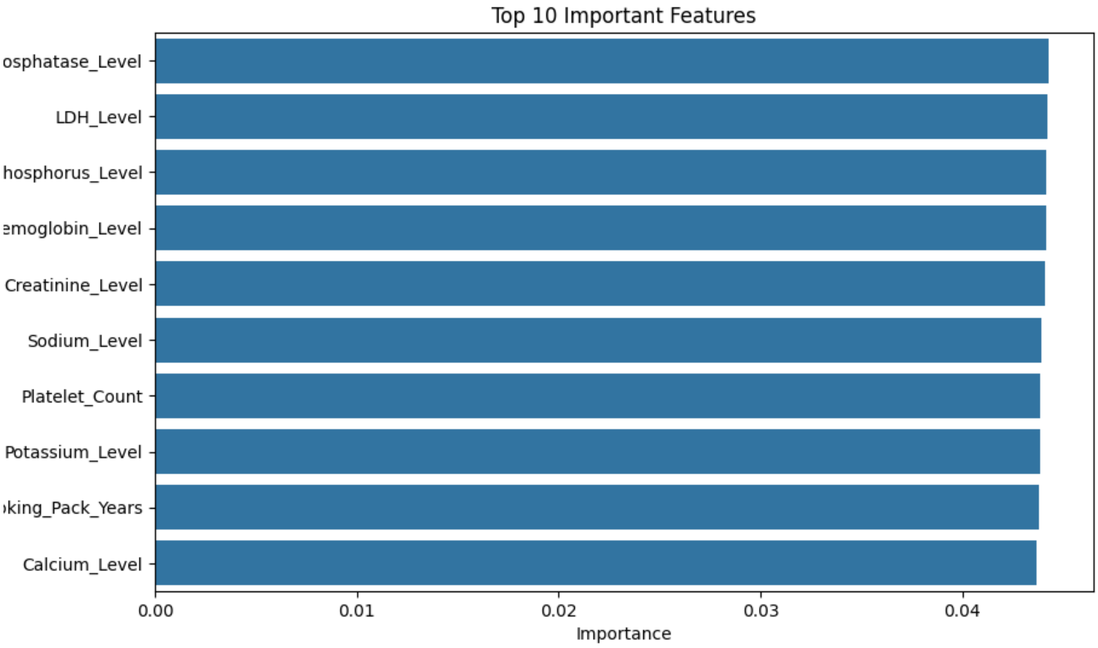

Figure 14: Top 10 important features

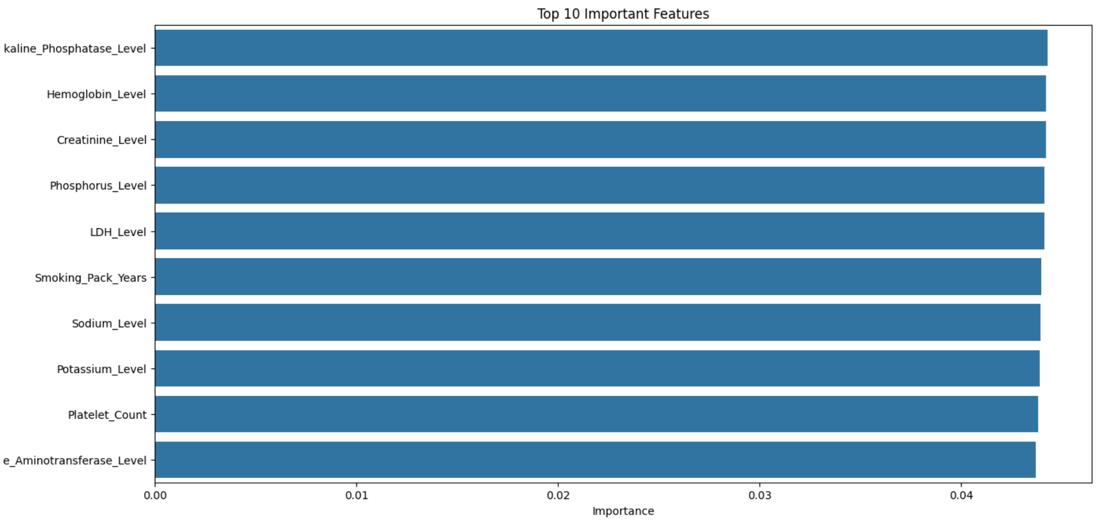

Figure 15: Updated top 10 important features

The most important features include:

- Alkaline Phosphatase Level
- LDH Level
- Phosphorus Level
- Hemoglobin Level

- Creatinine Level
- Sodium Level
- Platelet Count
- Potassium Level
- Smoking Pack Years
- Calcium Level

These features are mostly laboratory and clinical variables. This suggests that laboratory measurements may contain useful information for risk classification.

14 Feature Selection

Feature selection was applied to compare the performance of all features and only the top 10 important features.

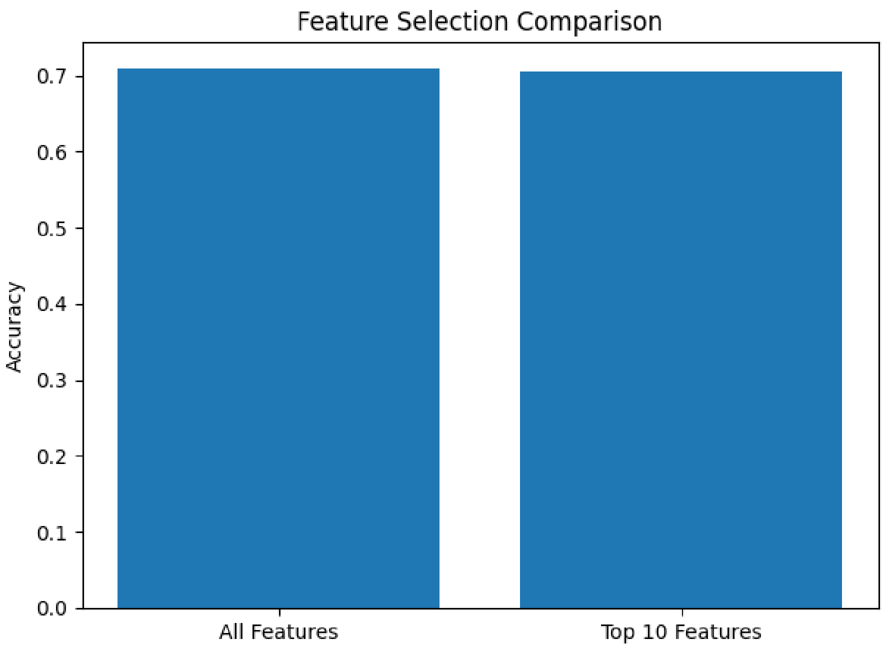

Figure 16: Feature selection comparison

Table 6: Feature Selection Result

Feature Set	Accuracy
All Features	0.71
Top 10 Features	0.70

The result shows that using only the top 10 features produced nearly the same accuracy as using all features. This means that the model can be simplified without losing much performance.

15 Training Time Comparison

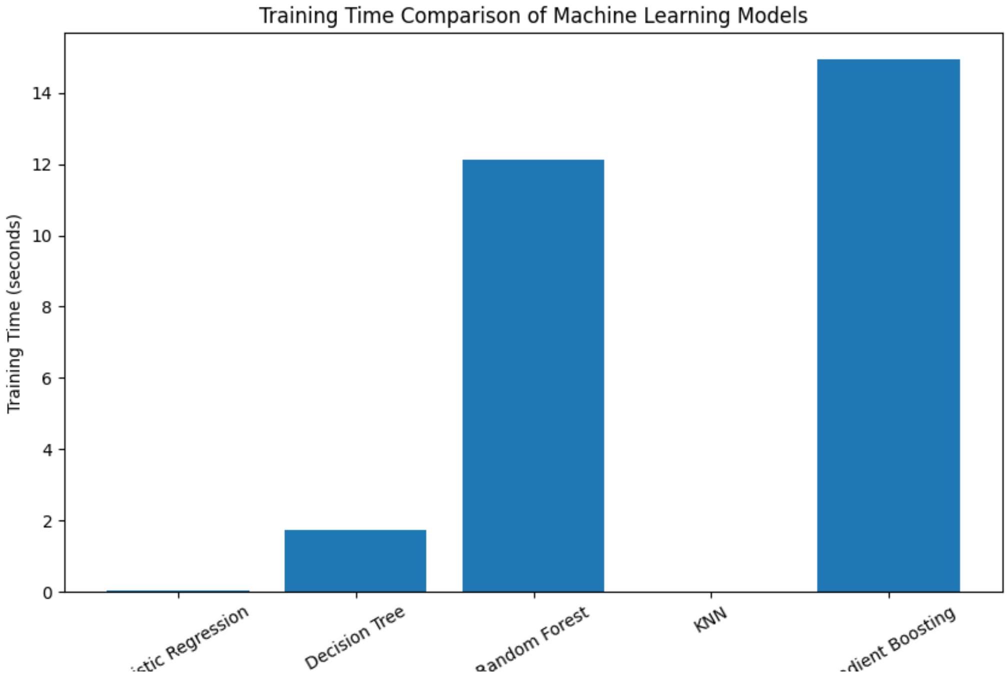

Figure 17: Training time comparison of machine learning models

The training time comparison shows that Random Forest and Gradient Boosting require more training time than simpler models. Logistic Regression and KNN are faster. However, training time should be interpreted together with model performance.

Table 7: General Model Comparison

Model	Accuracy	General Comment
Logistic Regression	0.71	Fast and simple baseline model
Decision Tree	0.58	Lowest accuracy, may overfit or underperform
Random Forest	0.71	Strong practical model with feature importance
KNN	0.64	Moderate performance, sensitive to scaling
Gradient Boosting	0.71	Strong accuracy but higher training cost

16 Graphical User Interface

A simple graphical user interface was developed to demonstrate how the model could be used in practice.

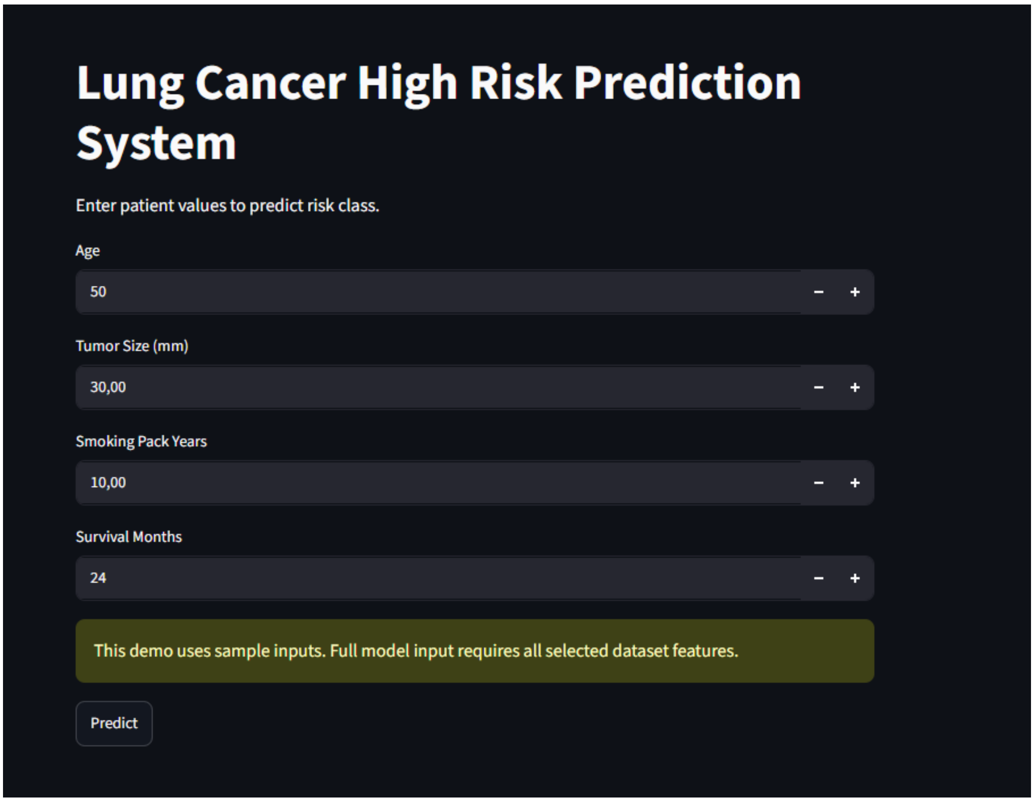

Figure 18: Lung Cancer High Risk Prediction System GUI

The GUI allows the user to enter sample patient values such as:

- Age
- Tumor Size
- Smoking Pack Years
- Survival Months

The GUI is a prototype. The full model requires all selected dataset features for more complete prediction.

17 Limitations

This project has several limitations:

- The target variable was created manually using stage and performance status.
- ROC-AUC values are close to 0.50, which indicates weak class separation.
- The dataset has class imbalance.

- The model should not be used as a real medical diagnosis system.
- More clinical validation is required.
- Hyperparameter tuning was limited.

18 Conclusion

This project developed a complete machine learning pipeline for high-risk lung cancer patient classification. The work included target creation, preprocessing, exploratory data analysis, model training, class imbalance handling, feature importance analysis, feature selection, training time comparison, ROC curve analysis, and GUI development.

The best practical model was selected as Random Forest because it achieved high accuracy, provided feature importance, and can capture nonlinear relationships. Logistic Regression and Gradient Boosting also achieved similar accuracy values.

The project demonstrates that machine learning can be useful for clinical risk classification. However, because ROC-AUC scores were close to 0.50, the model should be interpreted carefully. The system is suitable as an educational machine learning project and prototype, but not as a real clinical diagnostic tool.

19 Future Work

Future improvements may include:

- Hyperparameter optimization using GridSearchCV or RandomizedSearchCV.
- Testing additional models such as XGBoost, LightGBM, and CatBoost.
- Using cross-validation for more reliable results.
- Applying different imbalance methods such as ADASYN or class weighting.
- Improving feature engineering.
- Testing the model on real-world clinical datasets.
- Improving the GUI with all model input features.